

Basic Biotechnology

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CHAPTER 1 : The Biotechnology Century and It's Workforce

version one

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References :
INTRODUCTION TO BIOTECHNOLOGY THIRD EDITION
and Dr. Qutaiba notes and slides

1.1 what is biotechnology and what does it mean to you?

Introduction:

Can you imagine a world free of serious diseases, where food is abundant for everyone and the environment is free of pollution? These scenarios are exactly what many people in the biotechnology industry envision as they dictate their lives to this existing science.

Biotechnology is the science of using living organisms, or their products for human benefit to make a product or to solve a problem.

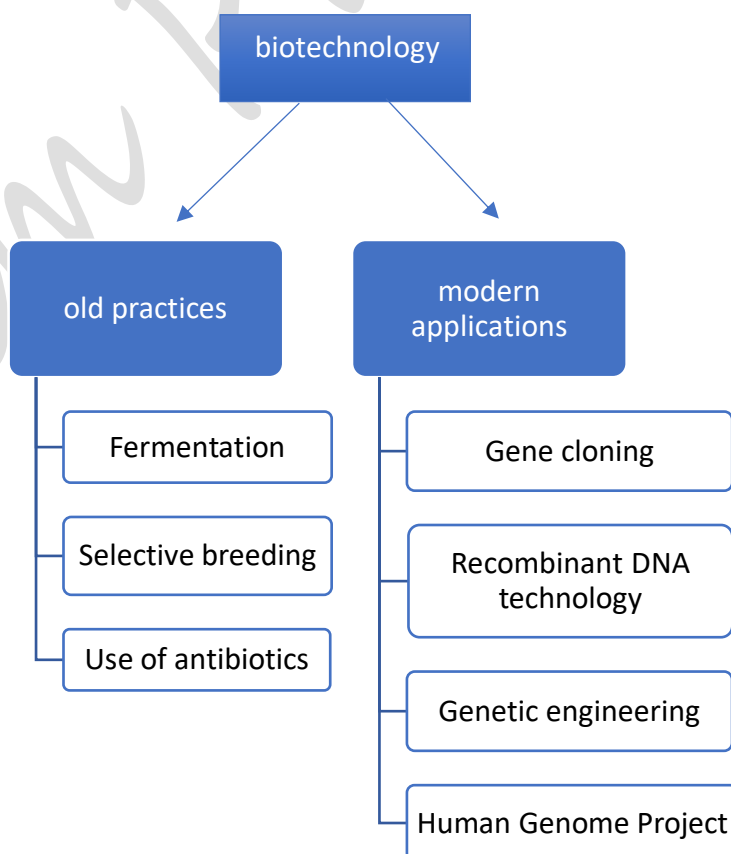
→ The benefit could be direct, such as using drugs to treat certain diseases or indirect. For instance, cleaning the environment and eliminate pollution which will lead in humans having a healthy life by breathing clean air, eating a clean food, and drinking a clean water free of contaminants and toxic materials.

You would be correct in thinking that biotechnology is a relatively new discipline that is only recently getting a lot of attention; however, it may surprise you to know that in many ways this science involves several ancient practices.

As we discuss in the next section, old and new practices in biotechnology make this field one of the most rapidly changing and exciting areas of science. It affects our everyday lives and will become even more important during this century – what some have called the “century of biotechnology”.

The focus of this course is on the modern applications of biotechnology.

A brief history of biotechnology



Although DNA cloning and the genetic manipulation of organisms are exciting modern-day techniques, **biotechnology is not a new science**. In fact, many applications represent old practices with new methodologies. Humans have been using biological organisms for their benefit in many processes for several thousand years.

Old biotechnology practices

- 1- **Fermentation:** using of microorganisms (yeast or bacteria) in food industry to make cheese, bread, pizza, yogurts, pickles, and alcoholic beverages such as beer and wine. During fermentation, some strains of yeast decompose sugars to derive energy, and in the process, they produce ethanol (alcohol) as a waste product.

Q) What is the type of yeast that is widely used to make bread or pizza ?

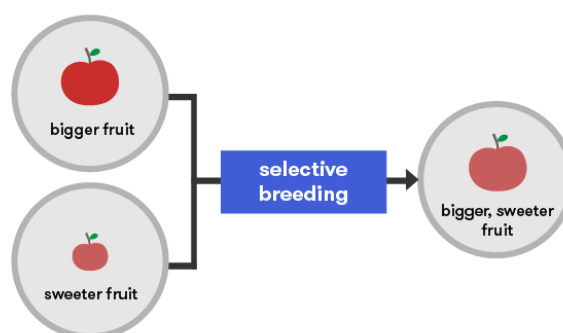
✓ saccharomyces cerevisiae, commonly called baker's yeast.

- 2- **Selective breeding:** organisms (crops or livestock) with desirable features are purposely mated to produce offspring with the same desirable characteristics.

- **Desirable features in Livestock** → highly produced milk and meat, vigor (physical strength and good health), producing a large number of offspring.
- **Desirable features in crops** → resistance to be damaged by insects, high yield, the ability to grow in harsh environmental conditions. For example, growing at low temperature or growing in a salty soil. Also, producing large number of offspring.
- Humans by selective breeding have been selecting certain types of genes that are responsible for a desirable phenotypes.
- Many of the fruits and vegetables that we are consuming in our daily basis didn't exist hundreds of years ago and these were produced by selective breeding between different types of plants.

Q) What is the difference between selective breeding and natural selection ?

Natural selection and selective breeding can both cause changes in animals and plants. The difference between the two is that natural selection happens naturally, but selective breeding only occurs when humans intervene. For this reason, selective breeding **is sometimes called artificial selection**.



3- Use of antibiotics: one of the most widespread and commonly understood applications of biotechnology, antibiotics are produced naturally by certain strains of bacteria and fungi, and they are used to treat infectious diseases → scientists have found ways to synthesize antibiotics chemically. Finding new antibiotics, new types of bacteria and fungi, is an active area for research.

- In 1928, Alexander Fleming discovered that the mold *Penicillium* inhibited the growth of a bacterium called *Staphylococcus aureus*, which cause skin disease in humans. Subsequent work by Fleming lead to the discovery and purification of the **antibiotic penicillin**.
- Antibiotics are substances produced by microorganisms that will inhibit the growth of other microorganisms.
- In 1940s, penicillin became widely available for medicinal use to treat bacterial infections in humans.

Note: although these examples are called old biotechnology practices they are still being used.

One of the applications of selective breeding is producing animal models for research.

“Casper” zebrafish, produced by selective breeding, is an important animal model for studying fish, stem cells, and cancer

- ❖ Casper was created by mating a zebrafish mutant that lacked reflective pigment with a zebrafish that lacked black pigment.
- ❖ Casper has already proven to be valuable for studying how cancer cells spread (**metastasis**).
- ❖ To study metastasis → scientists inject fluorescent tumor cells into the fish’s abdominal cavity and be able to track the migration of those cells to specific locations in the body.

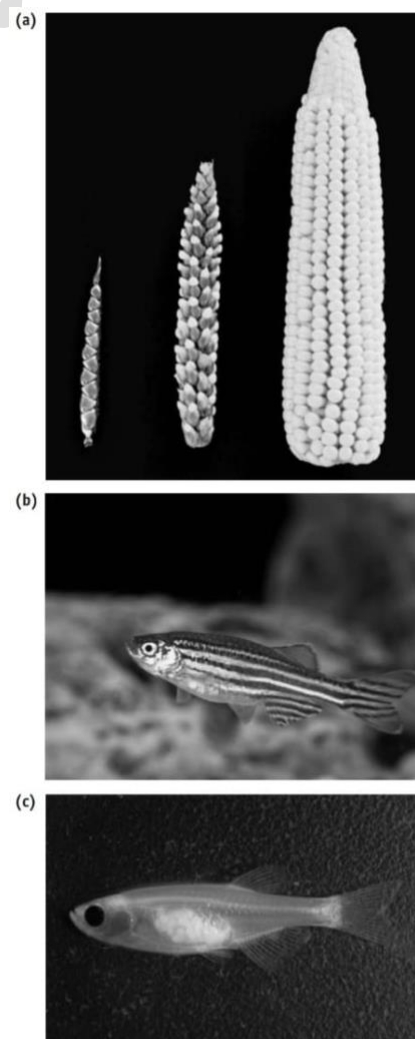


FIGURE 1.1 Selective Breeding Is an Old Example of Biotechnology That Is Still Common Today (a) Corn grown by selective breeding. From left to right is teosinte (*Zea canina*), selectively bred hybrids, and modern corn (*Zea mays*). (b) A normal zebrafish and (c) “Casper,” a transparent zebrafish produced by selective breeding.

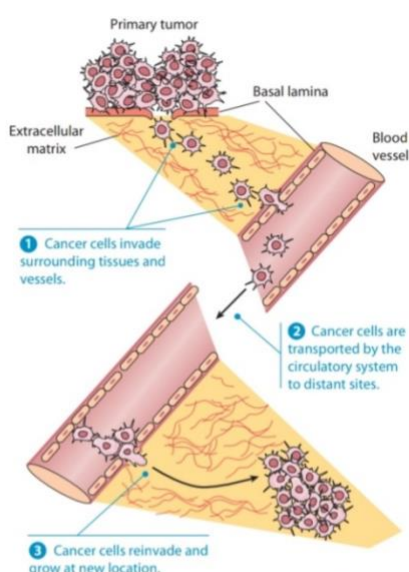


FIGURE 24-4 Stages in the Process of Metastasis. Only a small fraction of the cells in a typical cancer successfully carry out all three steps involved in metastasis: ① invasion into surrounding tissues and vessels, ② transportation via the circulatory system, and ③ reinvansion and growth at a distant site.

dairy cow that has a high milk production



strong bull that has high meat production



offspring will have a high milk and meat production and vigor

أول شي لازم تعرف انه تخصصنا قديم والبشر كانوا يمارسوه على اختلاف حضاراتهم منهم اليونان والرومان والفراعنة والصينيين من آلاف السنين مش من يوم ويومين وأقدم اثبات في الآثار الي تم اكتشافها يعود لعام 2000 قبل الميلاد يعني قديم كثير متخيل قديش؟
التقانات الحيوية هي بكل بساطه استخدام الكائنات الحية وتسخيرها لفائدة الإنسان إما لحل مشكله معينة او لصنع منتج معين.

مثال على المشكلة الي حليناها بواسطة تخصصنا هي مكافحة الأمراض المعدية - الي بتسببها الكائنات الحية الدقيقة و اسمها دقيقة لأنها لا ترى بالعين المجردة و ما بنقدر نشوفها إلا بالمجهر - عن طريق المضادات الحيوية و المطاعيم , مثال آخر هو التخفيف من التلوث و مثال بسيط عليه تلوث البحار و المحيطات الناتج عن تسرب النفط من البواخر الناقلة له , النفط عبارة عن زيت و ماده سامه للكائن الحي و الزيت ما بذوب في المي و اذا ما قدرنا نحل المشكلة رح تموت انواع كثير من الأسماك و الكائنات البحرية لهيك من احد تطبيقات التقانات الحيوية هي استخدام بكتيريا معينه قادرة على تحليل النفط الى مركبات ابسط و اخف سمية و بالتالي القضاء على التلوث الي حصل.

التقانات الحيوية بين الماضي والحاضر:

الماضي ← أولاً: عملية التخمر ومن خلالها بنقدر نصنع كثير من الأغذية مثل الجبنة والخبز والمخللات والمشروبات المش كويسه وبنعتمد فيها على البكتيريا والخميرة يعني لولاهم ما كان في عنا كل هذول الأكلات الي ذكرتهم

ثانياً: التلقيح الانتقائي وهو عبارته عن اختيار الانسان الكائنات الي بده اياها تتزاوج بناءً على صفات مرغوبه عشان تنتج جيل بحمل نفس هاي الصفات المرغوبه في النباتات وحيوانات المزرعة مثل الدجاج والبقرة والخرفان ومن الأمثلة على ذلك انه كانوا البشر زمان يختاروا القمح الي بقدر يعيش عند درجات الحرارة المنخفضة ويلقحوه مع قمح فيه صفه ثانيه مرغوبه مثل الانتاج العالي High yield يعني ينتج كميات كبيرة من البذور ليحصلوا على جيل بحمل الصفتين مع بعض.

ثالثاً: استخدام المضادات الحيوية لمواجهة الأمراض المعدية الي بتسببها البكتيريا وأول مضاد حيوي تم اكتشافه هو البنسلين عن طريق ألكسندر فليمنغ.

❖ كاسبر

- سمكة الكاسبر تم انتاجها عن طريق تلقيح انتقائي selective breeding
- سمكة الكاسبر تتميز انه جسمها شفاف لذلك بنقدر من خلالها ندرس كيف الخلايا السرطانية تنتشر بالجسم.
- تعتبر نموذج حيواني لدراسة الأسماك بشكل عام بالإضافة للخلايا الجذعية

Biotechnology: A science of many disciplines

Although a major focus of biotechnology involves molecular biology techniques, biotechnology is not a single, narrow discipline of study, it is an expansive, interdisciplinary field that absolutely relies on biology, chemistry, mathematics, computer science, and engineering in addition to other disciplines such as philosophy and economics.

Figure 1.3 provides a diagrammatic view of the many disciplines that contribute to biotechnology.

- Roots are primarily formed by work in the basic sciences research into fundamental processes of living organisms at the biochemical, molecular, and genetic levels. **When pieced together, with the help of computer science** → can lead to genetic engineering approaches.
- At the top of the tree, applications of genetic engineering can be put to work to create a product or process to help humans or our living environment → many of these processes have yet to be developed and await the intuitive participation of people working in biotechnology today.
- **Bioinformatics:** the use of computer science to analyze biological data.

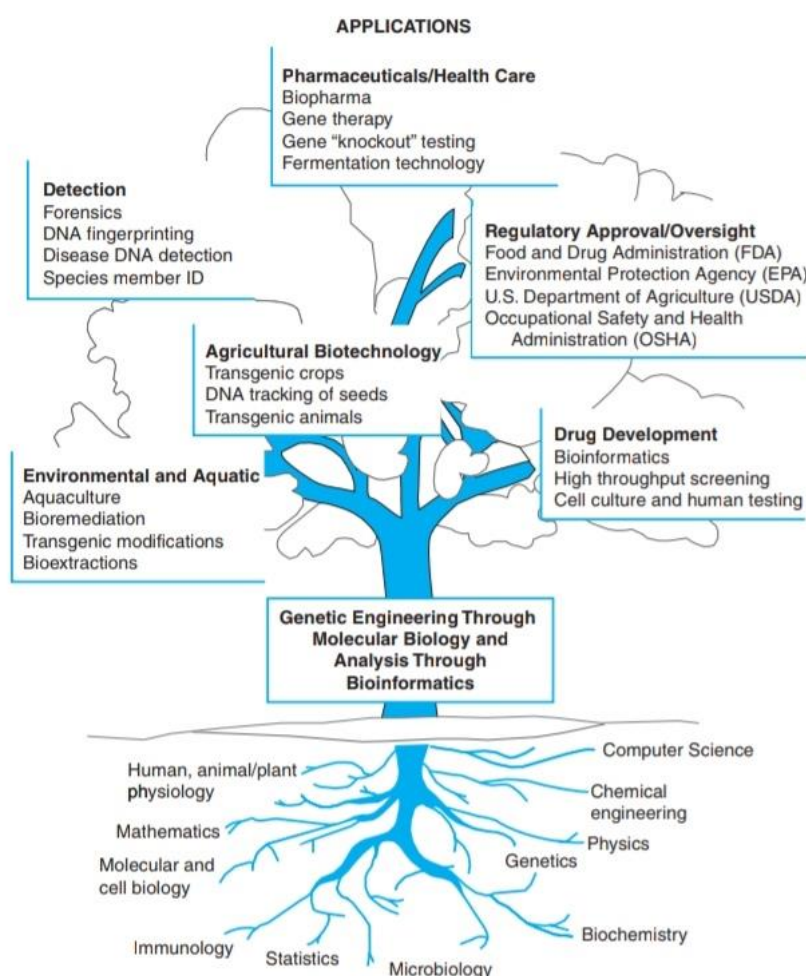
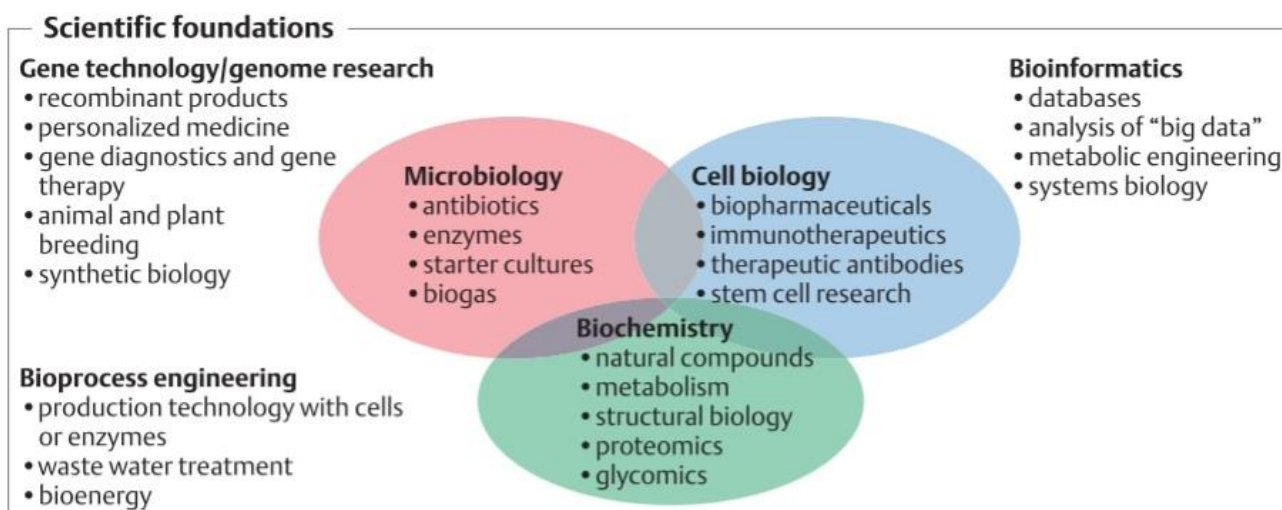


FIGURE 1.3 The Biotechnology Tree: Different Disciplines Contribute to Biotechnology The basic sciences are the foundation or "roots" of all aspects of biotechnology. The central focus or "trunk" for most biotechnological applications is genetic engineering. Branches of the tree represent different organisms, technologies, and applications that "stem" from genetic engineering and bioinformatics, central aspects of most biotechnological approaches. Regulation of biotechnology occurs through governmental agencies like the FDA, USDA, EPA, and OSHA (see Chapter 12).



Modern biotechnology applications

Since the 1960s, rapid development of our understanding of genetics and molecular biology has led to exciting new innovations and applications in biotechnology. As we have begun to unravel the secrets of DNA structure and function, new technologies have led to **gene cloning**, the ability to identify and reproduce a gene of interest in order to determine its sequence, study it, or express it, and **genetic engineering**, manipulating the DNA of an organism. Through genetic engineering, scientists are able to combine DNA from different sources. This process, called **recombinant DNA technology**, is used to produce many proteins of medical importance, including insulin, human growth hormone, and blood-clotting factors.

From its inception, recombinant DNA technology has dominated many important areas of biotechnology, and as you will soon learn, many credit recombinant DNA technology with starting modern biotechnology as an industry.

Recombinant DNA technology applications →

- 1- The development of disease-resistant plants.
- 2- Food crops that produce greater yields.
- 3- “golden rice” engineered to be more nutritious.
- 4- Genetically engineered bacteria capable of degrading environmental pollutants.

The **Human Genome Project** was an international effort that began in 1990 and ended in 2003 with the complete reference sequence of human DNA as well as the identification of thousands of genes and their location on chromosomes, association of certain genes with certain diseases.

Human Genome: the whole DNA content of the human cell that is distributed among 24 different chromosomes (22 autosomes and the X and Y chromosomes).

A scheme illustrating the events that lead to the discovery of modern biotechnology:

scientists discovered that the DNA is the genetic material → Watson and Crick provided a model for the DNA structure → the discovery and understanding of the DNA replication → new technologies were developed to manipulate the DNA → genetically modified organisms, new proteins, and drugs were produced as a result.

Q) How did the Human Genome Project accelerate the innovation in modern biotechnology ?

The big data that was produced by this enormous project has helped many companies to come up with a new biotechnology applications.

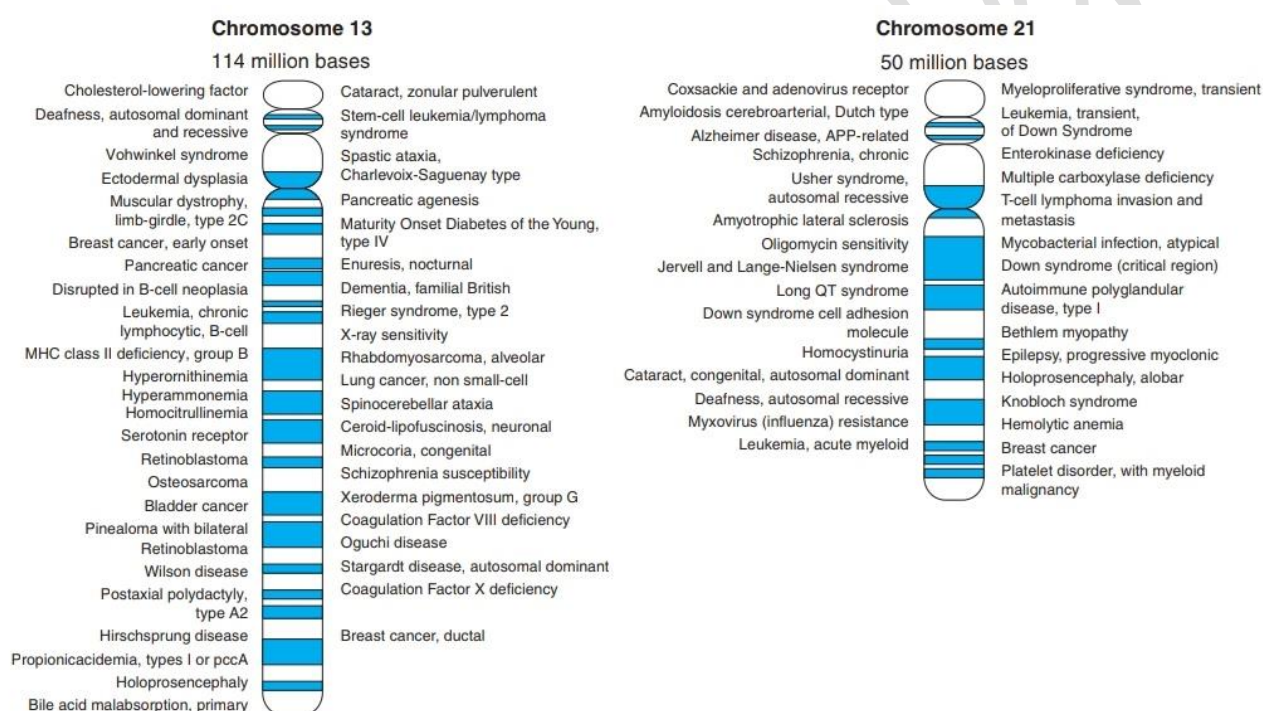


FIGURE 1.2 Gene Maps of Chromosomes 13 and 21 The Human Genome Project has led to the identification of nearly all human genes and has mapped their location on each chromosome. The maps of chromosomes 13 and 21 indicate genes known to be involved in human genetic disease conditions. Identifying these genes is an important first step toward developing treatments for many genetic diseases.

Products of modern biotechnology

- ❖ In fact, more than 65 percent of biotechnology companies in the united states are involved in pharmaceutical production.
- ❖ In 1982, the California biotechnology company **Genentech**, widely regarded as the world's first biotechnology company, received approval for recombinant insulin **Humulin**, used for the treatment of diabetes, as the first biotechnology product for human benefit (Figure 1.4).
- ❖ **Humulin was the first commercial modern biotechnology product.**

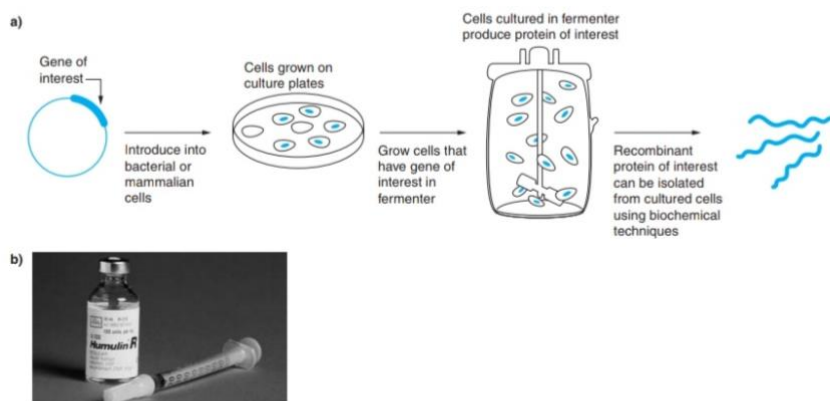


FIGURE 1.4 Using Genetically Modified Cultured Cells to Make a Protein of Interest Genes of interest can be introduced into bacterial or mammalian cells. Such cells can be grown using cell culture techniques. Recombinant proteins isolated from these cells are used in hundreds of different biotechnology applications. In this example, mammalian cells are shown, but this process is also commonly carried out using bacteria.

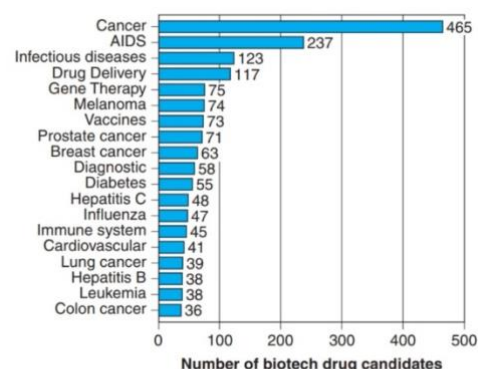


FIGURE 1.5 Investigational Biotechnology Drugs by Disease Category The production of drugs to combat cancer dominates the biotechnology industry's interest in treating human disease, with AIDS-related research and treatment of infectious diseases such as flu also near the top of this list.

- ❖ There are now several hundred drugs, vaccines, and diagnostic on the market with more than 300 biotechnology medicines in development targeting over 200 diseases.
- ❖ As shown in Figure 1.5, drug development by the biotechnology industry is focused on combining major diseases that affect humans, and over half of the new drugs in the development “pipeline” are designed to treat cancer.
- ❖ Table 1.1 provides a brief list of some of the top-selling biotechnology drugs and the companies that developed them.
- ❖ Diagnosis and/or treatment of a variety of human diseases and disorders → including acquired immunodeficiency syndrome (AIDS), stroke, diabetes, and cancer make up the bulk of biotechnology products on the market.
- ❖ **Many of the most widely used products of biotechnology are proteins created by gene cloning (Table 1.2). these proteins are called recombinant proteins → because they are produced by gene-cloning techniques.**

TABLE 1.1 *2016—Top 10 Biotechnology Drugs (Each with Worldwide Sales over \$5 Billion)			
Drug Name	Developer	Drug Type	Function (Treatment of Human Disease Conditions)
Humira	AbbVie	Antibody (monoclonal)	Rheumatoid arthritis, Crohn's disease, Ulcerative colitis
Harvoni	Gilead Sciences	Small molecule	Hepatitis C
Rituxan	Roche	Antibody (monoclonal)	Non-Hodgkin's lymphoma
Revlimid	Celgene	Small molecule	Multiple myeloma
Avastin	Roche	Antibody (monoclonal)	Colorectal cancer; breast cancer; non-small cell lung cancer; ovarian, brain, and cervical cancer
Herceptin	Roche	Antibody (monoclonal)	Breast cancer, gastric cancer
Enbrel	Amgen	Recombinant protein	Rheumatoid arthritis, psoriasis
Prevnar 13	Pfizer	Vaccine	Pneumococcal (<i>Streptococcus Pneumoniae</i>) antibacterial vaccine
Lantus	Sanofi	Peptide	Diabetes mellitus types I and II
Neulasta	Amgen	Recombinant protein	Anemia (neutropenia/leukopenia)

*Data based on the most recent source available at the time of publication: Morrison C, Lähteenmäki R. Public biotech in 2016—the numbers. *Nat Biotechnol.* 2017;35:623–629.

TABLE 1.2 EXAMPLES OF PROTEINS MANUFACTURED FROM CLONED GENES	
Product	Application
Blood factor VIII (clotting factor)	Treat hemophilia
Epidermal growth factor	Stimulate antibody production in patients with immune system disorders
Growth hormone	Correct pituitary deficiencies and short stature in humans; other forms are used in cows to increase milk production
Insulin	Treat diabetes
Interferons	Treat cancer and viral infections
Interleukins	Treat cancer and stimulate antibody production
Monoclonal antibodies	Diagnose and treat a variety of diseases including arthritis and cancer
Tissue plasminogen activator	Treat heart attacks and stroke

التقانات الحيوية: العلم المبني على العديد من المعارف

صحيح انه التركيز الأكبر عنا بالتخصص على تقنيات البيولوجيا الجزيئية لكن هو علم واسع ويعتمد على الكثير من العلوم الأساسية مثل الكيمياء والفيزياء والأحياء الدقيقة والوراثة بالإضافة لعلم الحاسوب وغيرها الموضحة بالشجرة وتمثل هذه العلوم الجذور التي يبنى عليها علم التقانات الحيوية.

بإجراء البحوث في هذه العلوم (يعني بس تكون متمكن منها كويس) وتدعيمها بالهندسة الوراثية (التعديل على DNA) وتحلل معلومات شغلك في المختبر (موقع الجين واسمه وتسلسله على طول الكروموسوم، عدد النيوكليوتيدات، شو التعديل الي بدك تعمله عليه ... الخ) باستخدام الحاسوب وتسمى هاي العملية bioinformatics والناج من كل شغلك بمثل الثمرة في الشجرة وهو النتيجة الي حصلت عليها ممكن تكون: علاج معين، مطعوم، حيوان، أو نبات معدل وراثياً.

التطبيقات الحديثة للتقانات الحيوية:

- 1- استنساخ الجينات ← لدراستها، تحديد تسلسلها، انتاج بروتيناتها
- 2- الهندسة الوراثية ← التلاعب بال DNA عن طريق تغيير تسلسل النيوكليوتيدات المكونة له.
- 3- **recombinant DNA technology** و هي تشكيل جزيء DNA هجين من مصادر مختلفة (مثال على ذلك لصق جين للإنسان في بلازميد او كروموسوم لبكتيريا)
- 4- مشروع الجينوم البشري.

منتجات التقانات الحيوية:

- معظم الشركات مختصة في تصنيع الأدوية.
- معظم الأدوية عبارة عن بروتينات تم تصنيعها بواسطة استنساخ الجينات (في شابتر كامل عنها بشرح العملية بالتفصيل الممل).
- أكثر من 300 دواء ومطعوم لعلاج ما يقارب 200 مرض منهم الإيدز والسكري والسرطان والسكتات الدماغية.
- نصف الأدوية التي في مرحلة التصنيع والتجريب (لم تطرح في الأسواق بعد) لعلاج مرض السرطان.

1.2 Types of Biotechnology

- ❖ Microbial Biotechnology → the use of microbes to produce food products (wine, cheese...etc.) or to produce drugs.
- ❖ Agricultural Biotechnology → plants can be engineered to
 - 1- be resistant to get damaged by insects
 - 2- to produce higher protein or vitamin content
 - agriculture waste can be used as a fuel.
- ❖ Animal Biotechnology → animals can be used as “bioreactors” to produce important product.
 - Scientists can create female transgenic animals that can express therapeutic proteins in their milk.
- ❖ Forensic Biotechnology → DNA fingerprinting (a collection of methods for detecting an organism’s unique DNA pattern) is a primary tool used in forensic biotechnology.
- ❖ Bioremediation → the use of biotechnology to degrade a variety of natural substances that contribute to environmental pollution.

- ❖ Aquatic Biotechnology → aquaculture is one of the oldest applications of aquatic biotechnology, that is raising finfish or shellfish in controlled conditions for use as food sources.
- ❖ Medical Biotechnology → stem cell technologies and gene therapy.
- ❖ Regulatory Biotechnology

1.3 Biological Challenges of the Twenty-First Century

Single nucleotide polymorphisms (SNPs) are single nucleotide changes or mutations in DNA sequence that vary from individual to individual

- ❖ These subtle changes represent one of the most common examples of genetic variation in humans.
- ❖ The sequence of the nuclear genome (3.3 billion base pairs) is approximately 99.5% identical between any two unrelated individuals → the remained 0.5% of the DNA is sequence differences is considered a genetic variation responsible for the phenotypic variation among individuals → sequence differences are SNPs, insertion-deletion polymorphisms, inversion polymorphisms, and copy number variants.
- ❖ On average, there is 1 SNP for every 1000 bp.
- ❖ For any two unrelated individuals, their genomes will differ in at least 5-7 million single nucleotide → each individual has his own unique combination of SNPs.
- ❖ The distribution of SNPs is uneven around the genome; many more SNPs are found in noncoding parts of the genome, in introns and in sequences that are some distance from known genes.
- ❖ If the SNP is found within the coding sequence it may or may not affect the encoded protein because the genetic code is redundant → most of the amino acids can be encoded by more than one codon.
- ❖ SNPs within the regulatory elements (enhancers & promoters) may alter the level of gene expression.
- ❖ The significance for health of the vast majority of SNPs is unknown and is the subject of ongoing research.
- ❖ Some SNPs are the cause of certain genetic diseases, such as sickle cell anemia.
- ❖ Most scientists believe that SNPs will help them identify some of the genes involved in medical conditions such as arthritis, stroke, cancer, heart disease, diabetes, and behavioral and emotional illnesses as well as a host of other disorders.

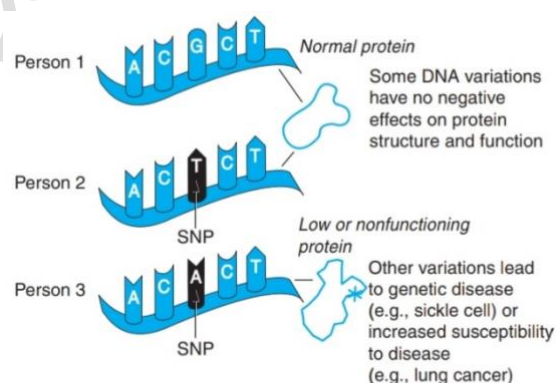


FIGURE 1.12 Single Nucleotide Polymorphisms A small piece of a gene sequence for three different individuals is represented. For simplicity, only one strand of a DNA molecule is shown. Notice how person 2 has a SNP in this gene, which has no effect on protein structure and function. Person 3, however, has a different SNP in the same gene. This subtle genetic change may affect how this person responds to a medical drug, or it may influence the likelihood that person 3 will develop a genetic disease.

- ❖ Some SNPs have been found to be inherited with specific alleles → as shown in the **Figure 1.12**, the normal allele of a gene would be associated with a specific SNP (T) whereas the mutant allele would be associated with the SNP (A).
- ❖ When SNPs cause mutations within the coding sequence of genes or associated with certain disease alleles, they can be used as **genetic markers** → instead of screening the whole allele to determine if it is normal or mutant, the **SNP associated with that allele is screened only**.
- ❖ The very small distance between the allele and its associated SNP guarantee that they will be inherited together → a crossing over event is very unlikely to separate them.
- ❖ **Locus (plural loci)** is a DNA segment that could be large and contain many genes or it may be a single gene; or it may even be just a single base in the genome.
- ❖ **Alleles** are alternative versions of the DNA sequence at a locus.

Q) what is the difference between SNPs and mutations ?

If there are two or more relatively common alleles (defined by having an allele frequency > 1%) at a locus in a population → that locus is said to exhibit **polymorphism** (literally “many forms”) in that population. If there is an allele having an allele frequency < 1% at a locus → **mutant allele**.

DNA Microarray (gene chip) are used to determine the combinations of SNPs within an individual

- 1- Isolate DNA from patient.
- 2- Apply this sample to a microarray which contains many DNA sequences.
- 3- Compare patterns of DNA binding between patient's DNA and DNA on microarray to reveal patient's SNP patterns.

Pharmacogenomics is customized medicine

- ❖ Tailor-designing drug therapy and treatment strategies based on the genetic profile of a patient based on the combination of SNPs he has.
- ❖ Do microarray analysis and then design drugs against genes that are mutated for an individual patient.
- ❖ Some SNPs can affect drug metabolism either by causing **side effects** or mask the **effectiveness of the drug** → that's why there is no drug that works on everyone and a same type of drug may have different responses from different patients.

Metabolomics: a snapshot of the small molecules produced during cellular metabolism

- ❖ Glucose, cholesterol, ATP, and signaling molecules.
- ❖ Can distinguish between disease process and physiological adaptation.
- ❖ Because each individual has his own unique combination of SNPs and some SNPs affect drug metabolism → some patients may require a change in the drug dose.
- ❖ The medical condition of a patient can be determined by analyzing the small molecules in the blood or within cells.
- ❖ **Normal individuals would have certain levels of these small molecules and these levels are altered in patients.**

Note: pharmacogenomics, metabolomics, proteomics and all sciences that their names end with (-omics) were developed by scientists depending on the big data produced by the Human Genome Project.

Nanotechnology

- ❖ Applications that incorporate extremely small devices (a “nano” scale).
- ❖ Small particles that can deliver drugs to cells.
- ❖ **Much better than chemotherapy in treating cancer** → chemotherapy drugs attack both normal cells (including neurons) and cancer cells; that’s why patients suffer from chemotherapy side effects.

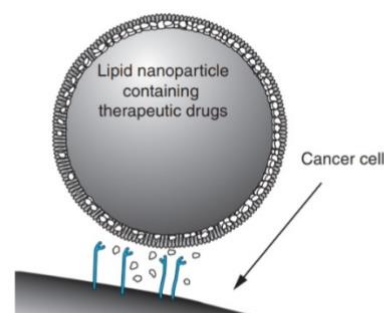


FIGURE 1.13 Nanobiotechnology in Action Nanoparticles containing chemotherapy agents can be specifically directed to target cancer cells by a coat of tumor-specific proteins bound to the target cells. In this way, chemotherapy agents that cannot pass through the cell membrane can be released inside these target cancer cells.

Gene therapy technology: replacing or augmenting defective genes with normal copies of the gene

- ❖ Some genetic diseases are caused because of mutations within the coding region (exons) of genes.
- ❖ Lipid vesicles or viruses are used to introduce the normal copy of the gene.
- ❖ Scientists are now focusing on gene editing tools such as **Crisper** more than gene therapy techniques.
- ❖ Still have barriers to overcome before this technology becomes safe and effective.
- ❖ Obstacles currently prevent gene therapy from being widely used in humans →
 - How can normal genes be delivered to virtually all cells in the body?
 - What are the long-term effects of introducing extra genes in humans?
 - What must be done to ensure the proper protein is made after the genes are delivered to the body?

Small interfering RNA (siRNA) is emerging technology to silence genes that are involved in disease progression

- ❖ siRNA inhibit gene expression by binding to the mRNA that is blocking its translation.
- ❖ In 2019, the first RNAi (RNA interference) drug named Patisiran, sold under the brand name Onpattro was approved by the FDA.
- ❖ Givosiran is another drug that has recently been approved by the FDA.

Stem cell technologies are expected to provide powerful tools for treating and curing disease

- ❖ Stem cells are immature cells that grow and divide to produce different cell types, such as skin, kidney, and blood cells.
- ❖ Most stem cells are from embryos called embryonic stem cells (ESCs) but they are controversial since the process involves death of an embryo.
- ❖ Some stem cells are from adult cells (ASCs).
- ❖ ESCs can give rise to many types of differentiated cells.
- ❖ Regenerative medicine → growing skin cells, blood cells, and even whole organs in the lab and using these to replace damaged tissue or failing organs such as the liver, pancreas, retina (Figure 1.14).
- ❖ Cell potency: the ability of the cell to differentiate into different cell types.
- ❖ Based on cell potency, stem cells can be classified into the following types → potency increases as we go from unipotent to totipotent.

- 1- Totipotent stem cells such as zygote (Highest potency).
- 2- Pluripotent.
- 3- Multipotent (adult stem cells).
- 4- Oligopotent.
- 5- Bipotent.
- 6- Unipotent (lowest potency).

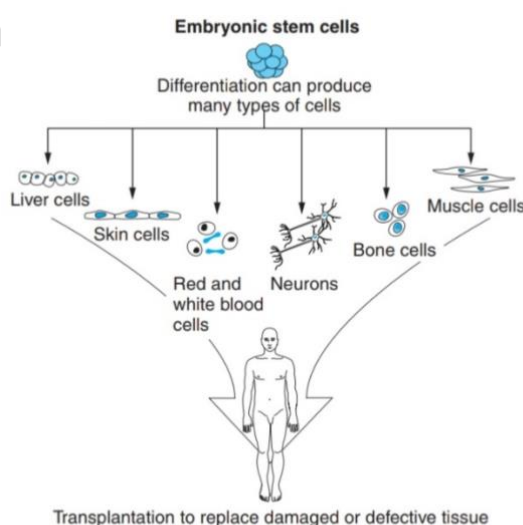


FIGURE 1.14 Embryonic Stem Cells Can Give Rise to Many Types of Differentiated Cells Embryonic stem cells (ESCs) are derived from embryos or early-stage fetuses; they are immature cells that can be stimulated to differentiate into a variety of cell types.

1.4 The Biotechnology Workforce

Biotechnology is a global industry

- ❖ Annual sales of biotherapeutics are in excess of \$225 billion worldwide and doubling about every 5 years!
- ❖ Many of the pharmaceutical companies produce biotechnology drugs.
- ❖ Differences between a biotech and pharmaceutical company:
 - Pharmaceutical companies is involved in drug development (chemically synthesizing or purifying compounds to make the drug)
 - Pharmaceutical company does not use living organisms to grow or produce a product
 - Biotech company is involved in drug development using live organisms
 - Biotech companies also create products that are not drugs



FIGURE 1.15 The Biotechnology Industry Provides Exciting Opportunities for Many Types of Scientists From biologists and chemists to engineers, information technologists, and salespeople, the biotechnology industry offers a great range of high-tech employment opportunities. Shown here is a senior undergraduate student working on a biotechnology research project. Gaining research experience as an undergraduate is an excellent way to prepare for a career in biotechnology.

Q) How does a biotech company start?

- Generally created as a startup company with few employees backed by venture capital investments (derived from financial institutions and private donors called angel investors)
- Why invest as an angel donor? To ultimately have company ownership in exchange for donating funds –get rich if company is successful!
- Takes 10 years to get a product to market at a big cost!
- Eventually startups are bought out by bigger established companies
- Bringing drug close to market = value to the company– File for initial public offering (IPO) stock

Jobs in Biotechnology

- Research and development: preclinical, discovery research, bioinformatics, lab safety.
- Operations: process/product development, manufacturing and production.
- Quality: quality control and assurance.
- Clinical research: clinical research and regulatory affairs.
- Finance and administration: finance, business development, administration, information systems, legal, facilities management.

TABLE 1.3 TOP FIVE BIOTECHNOLOGY COMPANIES AND TOP FIVE PHARMACEUTICAL COMPANIES BY REVENUE

Biotech Companies	Revenue (Millions)
Amgen	\$14,268
Genentech (now part of Roche)	\$11,724
Genzyme	\$ 3,187
UCB	\$ 3,169
Gilead	\$ 3,026
Pharma Companies	
Johnson & Johnson	\$61,897
Pfizer	\$50,009
Roche	\$49,051
GlaxoSmithKline	\$45,830
Novartis	\$44,267

Adapted from: Ernst and Young, *Beyond Borders: Global Biotechnology Report 2010* (www.ey.com/beyondborders). Revenue based on preliminary results reported by companies.

What is the best approach to landing a good job in a biotech company?

- 1- Obtain a degree in molecular biology, biochemistry, or bioinformatics
- 2- Minor in information technology
- 3- Take lots of math courses
- 4- Have great skills in oral and written communication
- 5- Must be a collaborative team player
- 6- Why go into this field? It pays well and is personally rewarding!

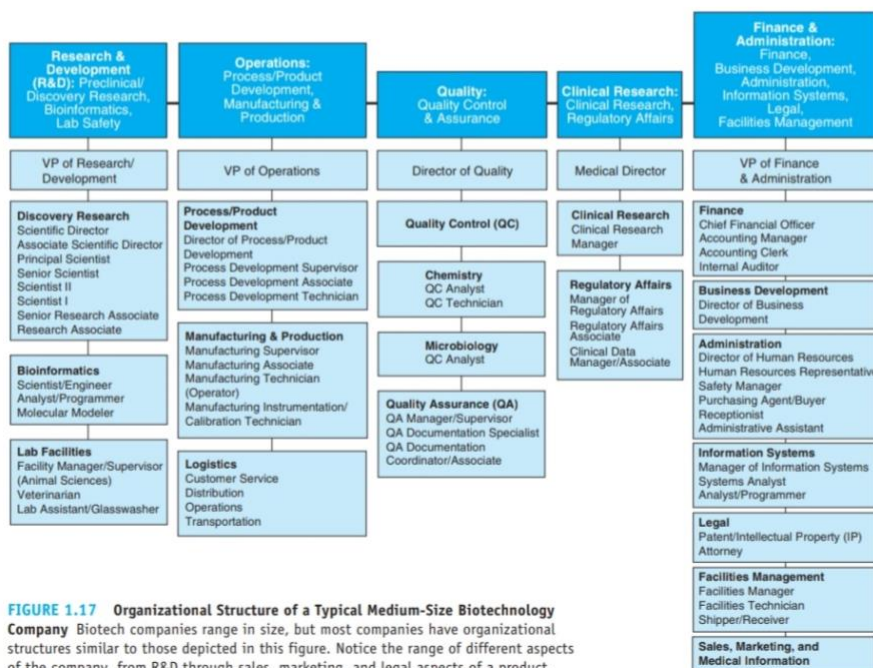


FIGURE 1.17 Organizational Structure of a Typical Medium-Size Biotechnology Company Biotech companies range in size, but most companies have organizational structures similar to those depicted in this figure. Notice the range of different aspects of the company, from R&D through sales, marketing, and legal aspects of a product.

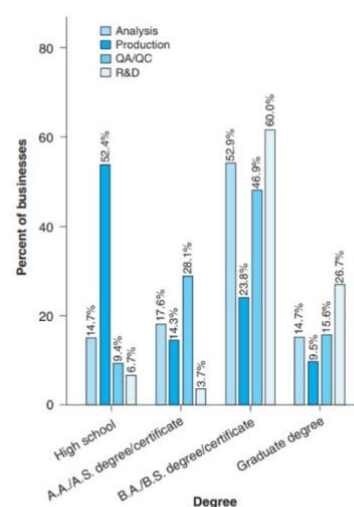


FIGURE 1.18 Minimal Level of Education Required of Entry-Level Technicians The Resource Group performed a survey of entry-level education requirements for 69 biotechnology companies in the three-county area of central coast California. Results are comparable with other areas of the United States and indicate that R&D generally requires a greater amount of training than other job areas.

ملاحظات مهمة:

هاي الجداول و الي قبل حطيتهم لتستفيد منهم فلا تبصمهم و بالنسبة لجدول أسماء الشركات و الأدوية اقرؤه قراءة و المهم هو انك تعرف انه في أنواع لهاي الأدوية (لقاحات، مضادات حيوية... الخ)

رسمة ال metastasis لا تحفظها.. حطيتها لتفهمها اكثر

هيك ان شاء الله يكون شطبتلك المادة كاملة % 99.9 من نوات الدكتور قتيبة موجوده بالشرح بالإضافة للشغلات المهمة بالكتاب .

شوف الأسئلة الي بالآخر مهمة جداً جداً

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) Scientists at the biotechnology company Genentech created the first recombinant DNA product for use in humans. Approved by the Food and Drug Administration in 1982, this product was:

- A) A gene chip
- B) Chymosin
- C) Insulin
- D) Erythropoietin
- E) Human growth hormone

2) Which area of biotechnology primarily involves using living organisms to process, degrade, and clean up environmental pollution?

- A) Bioinformatics
- B) Marine biotechnology
- C) Agricultural biotechnology
- D) Bioremediation
- E) Genomics

3) The use of computer hardware and software to analyze biological data such as DNA sequence comparison data is known as:

- A) Bioremediation
- B) Nanotechnology
- C) Bioinformatics
- D) Genomics
- E) Recombinant DNA technology

4) Studying the entire complement of DNA in an organism's cells is known as:

- A) Bioinformatics
- B) Recombinant DNA technology
- C) Genomics
- D) Bioremediation

E) Nanotechnology

5) Combining DNA from different sources is an example of:

A) Bioinformatics

B) Bioremediation

C) Recombinant DNA technology

D) Nanotechnology

E) Genomics

6) _____ is customized medicine based on a person's genetics.

A) Genomics

B) Pharmacogenomics

C) Bioremediation

D) Bioinformatics

E) Recombinant DNA technology

7) _____ involves farm-raising finfish, shellfish, algae, and other aquatic organisms.

A) Aquaculture

B) Transgenic technology

C) Bioremediation

D) Pharmacogenomics

E) Biopharming

8) _____ is the use of genes for treating human genetic disorders.

A) Genetics

B) Pharmacogenomics

C) Recombinant DNA technology

D) Genomics

E) Gene therapy

9) Selective breeding involves:

- A) Combining sperm and egg cells from different species to produce hybrid organisms
- B) Genetic engineering of animals and plants to improve growth characteristics
- C) Mating organisms with desirable characteristics
- D) The use of fermentation to produce biotechnology products
- E) None of these choices

10) _____ is the use of living organisms or their products to clean up the environment, improve human health and provide more nutritious and disease-free food.

- A) Biodegradation
- B) Bioprocessing
- C) Biotechnology
- D) Bioremediation
- E) Genetic engineering

11) Subtle differences in DNA sequences that vary from person to person are called:

- A) Genomics
- B) Single-nucleotide polymorphisms
- C) Pharmacogenomics
- D) Chromosomes
- E) DNA microarrays

1	2	3	4	5	6	7	8	9	10
c	d	c	c	c	b	a	e	c	c
11									
b									

**BASIC BIOTECHNOLOGY****BT230/BT232**

Quiz No.1 (2.10.19)

1. Which of the following types of cells has the most potency?
A. zygote
B. Liver cell
C. ESC
D. Adult stem cell
2. Combining DNA from different sources is an example of _____.
A. bioinformatics
B. recombinant DNA technology
C. nanotechnology
D. genomics
3. Match each term on the left with its appropriate description from the list to the right
 - Insulin → Recombinant protein,
 - Gene therapy → Replacing or augmenting defective genes
 - Pharmacogenomics → customized medicine,
 - siRNA → gene silencing
4. The process which uses microorganisms to make bread, cheeses, yogurts and alcoholic beverages is known as
A. Recombinant protein production
B. Natural selection
C. Fermentation
D. Genetic Engineering
5. Write the name the scientist who was the first to discover the antibiotic penicillin
 - **Alexander Fleming**
6. Fill in the blank
 - Stem cells are **immature** cells that grow and divide to produce different cell types. Most stem cells are from embryos called **ESCs**, while some types of stem cells are from adult cells (ASCs).

Best Wishes